

Luc Souilla

POST-DOCTORAL FELLOW

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About me

I am an interdisciplinary researcher driven by the main question: How will global warming shape human health across the lifespan? During my PhD, I developed strong expertise in the physiology and physical activity of child and adolescent health in clinical settings. Driven by a keen interest in climate change and strengthened through international collaborations, I began exploring the intersection of global warming and paediatric health, further deepening my focus on human adaptation to heat. I held a postdoctoral fellowship to investigate the physiological and behavioural responses to heat exposure. I intend to integrate my expertise in child health and environmental physiology to acquire novel methodologies aimed at understanding and mitigating the impact of heat on children's health at individual level with lab-based studies and population level using cohort-based studies.

Current position

Post-doctoral position — *Montreal, Canada*

MONTREAL HEART INSTITUTE - UNIVERSITY OF MONTRÉAL — *Nov. 2024*

- Physiological and perceptual responses to heat exposure
- Human Integrative Physiology Laboratory (Pr. Daniel Gagnon)

Education

PhD Fellow and Associate Lecturer — *Montpellier, France*

SPORT SCIENCE FACULTY — *Oct. 2021 - Sept. 2024*

- Evaluation and optimisation of physical fitness and physical activity in children with inherited cardiac diseases
- Supervised by Pr. Pascal Amedro, Pr.Jean-Luc Pasquie, Pr.Denis Mottet

MSc degree in Sport Sciences and Exercise — *Montpellier, France*

SPORT SCIENCE FACULTY — *Sep. 2020 - July. 2021*

- Sciences, Technology, Movement, top of the class.

BSc in Sport Sciences - Exercise and Health — *Toulouse and Montpellier, France*

SPORT SCIENCE FACULTY — *Sep. 2018 - July. 2019*

- Exercise and chronic conditions, top of the class.

Selected Publications

I have published 13 papers in peer-reviewed journal, 6th as first author and 1 as last author, with a total of 225 citations.

- Lefebvre, K., Dahhak, A., Pamart, N., […]**Souilla, L.** (2026). *When does sweating change during puberty? Revisiting the developmental timing of sex differences in sweating capacity*, The Journal of Physiology. **link here**
- Vanos, J.K., Guardaro, M., Middel, A., Ravanelli, N., […], **Souilla, L.**, […], Ouazaa, Y., Gagnon, D. (2026). *Hot Topics: Building Capacity for Extreme Heat Adaptation Through Transdisciplinary Training.* Bull. Amer. Meteor. Soc., **link here**
- Dahhak, A., Pamart, N., […]**Souilla, L.**, Deshayes, T. (2025). *Fight Fire with Fire: Is Heating Rather than Cooling the Key to Muscle Recovery?*, The Journal of Physiology. **link here**
- **Souilla, L.**, Shaheen, A., Nasser Mostafa, A., Abuzerr, S. (2025).*The Escalating Health Crisis in Gaza Amidst Armed Conflict and Heatwaves*, Global Health Action. **link here**.
- **Souilla, L.**, Werner, O., Huguet, H., […], Amedro, P., (2025). *Cardiopulmonary Fitness and Physical Activity Among Children and Adolescents With Inherited Cardiac Disease*, JAMA Open. **link here**.
- **Souilla, L.**, Guillaumont, S., Auer, A., […], Amedro, P., (2024). *Cardiac rehabilitation in children and adolescents with long QT syndrome: the RYTHMO'FIT pilot study*, BMC Sports Sci, Med and Rehab. **link here**.
- **Souilla, L.**, Amedro, P., Morrison, S., (2024). *Children with cardiac disease and heat exposure: catastrophic converging consequences?*, Pediatric Exercise Science. **link here**.

- **Souilla, L.**, Larsen, A.C., Juhl, C., Skou, S.T., Bricca, A., (2024). *Childhood and adolescence physical activity and multimorbidity later in life: A systematic review*, Journal of Multimorbidity and Comorbidity. **link here**.
- **Souilla, L.**, Avesani, M., […], Pasquie, J-L., Guillaumont, S., and Amedro, P., (2023). *Cardiorespiratory fitness, muscle fitness, and physical activity in children with long QT syndrome: A prospective controlled study*, Front. Cardiovasc. Med. **link here**.
- Amedro, P., Gavotto, A., Huguet, H., **Souilla, L.**, […], Guillaumont, S. (2024). *Early hybrid cardiac rehabilitation in congenital heart disease: the QUALIRE-HAB multicentre randomized controlled trial*, European Heart Journal. **link here**.
- Amedro, P., Matecki, S., […], **Souilla, L.**, […], Gavotto, A. (2023). *Reference Values of Cardiopulmonary Exercise Test Parameters in the Contemporary Paediatric Population*, Sports Medicine. **link here**.
- Gavotto, A., Mura, T., […], **Souilla, L.**, […], Amedro, P. (2023). *Reference values of aerobic fitness in the contemporary paediatric population*, European Journal of Preventive Cardiology. **link here**.
- Amedro, P., Werner, O., […], **Souilla, L.**, […], Pasquie, J-L. (2021). *Health-related quality of life and physical activity in children with inherited cardiac arrhythmia or inherited cardiomyopathy: the prospective multicentre controlled QUALIMYORYTHM study rationale, design and methods*, Health Qual Life Outcomes. **link here**.

Professional Experiences

Workshop facilitator
Montréal, Canada

Elementary and High School — *March. 2025*

- Fresk animation for children: Cardiovascular Thermoregulation in human and impact of global warming

Project manager
Montpellier, France

University of Sport Science — *Sept. 2023 - January. 2024*

- Kickoff event on Sports and Climate Change
- Organizing an awareness day on climate change and the impact of sports for students

Research Mobility
Odense, Denmark

Southern Denmark University - Department of Sport Sciences and Clinical Biomechanics — *May. 2023 - July. 2023*

- ERC projet MOBILIZE
- Effect of tailored exercise program on multimorbidity / Systematic review on physical activity, childhood and multimorbidity

Research intern
Montpellier, France

PhyMedExp - INSERM U1046 — *Sep. 2020 - July. 2021*

- Supervised by Pascal Amedro
- Cardiorespiratory Fitness and Muscular Deconditioning in congenital Long QT syndrome children

$ Competitive Grants

I have been involved in successful research proposals, securing €10,000 as the principal investigator and €1,117,100 as a collaborator.

Climate Fresk Welcome Day: Orientation Day on climate awareness for students of the exercise science department
September 2023

EXCELLENCE PROGRAM I-SITE from University of Montpellier — *€17,100*

- Contributed to the conception and design of workshops and activities for the event (co-applicant)

RYTHMO'FIT: Interventional pilot study
July 2022

From Saint Pierre Fundation — *€10,000*

- Designing exercise rehabilitation program for children with long QT syndrome (Principal investigator)

QUALIREHAB-ONCO: Effect of home based exercise rehabilitation program on childhood cancer survivors
June 2021

From ARC: Fundation for cancer research — *€450,000*

- Help to design and draft the intervention, final proofreading (co-applicant)

QUALINEUROREHAB: Neuro-cardiac Rehabilitation in Youth With Congenital Heart Disease
2021

European Joint Programme on rare diseases — *€650,000*

- Help to design and draft the intervention, final proofreading (co-applicant)

Fellowship & Awards

My research has been supported by over €261,473 in fellowships and awards.

FRQS PostDoctoral Fellowship
May 2026-May 2028

Fonds de recherche du Quebec — *56 500€*

- Four postdoctoral fellowship availables for 14 applicants (score: 87/100, rank 4/14)

CHIR PostDoctoral Fellowship
June 2026

Canadian Institutes of Health Research — *86 483€*

- (Postdoctoral fellowship awarded for two years support: 4.40/5, rank 77/408))

MSCA PostDoc Seal of Excellence
March 2026

European Commission – Horizon Europe — *Unfunded*

- Awarded the Seal of Excellence for a highly ranked MSCA Postdoctoral Fellowship application (scored 89.40/100, unfunded due to budget constraints)

PhD Scholarship
Oct.2021-2024

From University of Montpellier — *110 000€*

- PhD fellowship awarded through a competitive doctoral entrance exam for three-years support

International mobility EXPLORE #4
May-June 2023

EXCELLENCE PROGRAM I-SITE from University of Montpellier — *7990€*

- PhD mobility to Southern Danemark University under supervision of Prof. Søren T. Skou, working on ERC grant funded project: MOBILIZE

Call for support for national or international mobility
May 2021

From University of Montpellier — *500€*

- Support for participation in the International Congress of Association for European Paediatric Congenital Cardiology in Geneva, Switzerland

Media & Outreach

I have been involved in several media and outreach activities, including interviews, video creation, and public talks, to disseminate my research findings.

Live on radio 98.5FM
Montreal, Canada

Physiological Responses to Heat Stress and Implications for Health [Listen here] — *2026*

School of Global Health
Copenhagen, Danemark

In-situ responses of older adults living in social housing before, during and after a heatwave [Watch video] — *2025*

CBC/Radio Canada
Montreal, Canada

Experimental exposure to extreme heat by a journalist [Watch video] — *2025*

Cardiogen
Montpellier, France

Co-created public outreach video on exercise rehabilitation benefits for children with cardiac diseases [Watch video] — *2025*

Teaching Experience

I have accumulated 132 hours of teaching involving course design and teaching using active learning strategies, creating workshops and quizzes to engage students.

Courses	Level	Hours	Contribution
Exercise physiology and Cardiovascular Impairments	BSc	50h	Delivered with provided material
Anatomy of Human Movement	BSc	26h	Delivered with provided material
Climate change and challenges for physical activity and sports	MSc	16h	Created course content [Watch video]
Neuroscience of Motor Performance	BSc	16h	Delivered with provided material
Adaptations to Exercise and Health	BSc	16h	Delivered with provided material
Research Methodology	MSc	8h	Delivered with provided material
Exercise programs for children with inherited cardiac diseases	BSc	4h	Created course content

Supervision

Mentoring MSc Student
Bordeaux, France

Sports Science University — *Oct. 2023–Oct. 2024*

- Trained student in cardiopulmonary exercise testing and accelerometry; assisted in designing research protocols
- Supported protocol design, dissertation writing, and funding applications

Mentoring BSc Student

Montpellier, France

Sports Science University

Oct. 2023–Dec. 2023

- Mentored student during a kickoff event on Sports and Climate Change
- Provided individual support for developing personal projects

Mentoring MSc Student

Bordeaux, France

University Hospital of Paediatric Cardiology

Jan. 2023

- Assisted in designing exercise interventions; supported dissertation writing and funding applications

Conferences & Presentations

Annual research days: School of Kinesiology and Physical Activity Sciences

Montréal, Canada

Poster Presentation

March 2026

- Souilla L. (2026). In situ responses of elderly people living in social housing before, during, and after a heatwave.

Global Health Symposium: Extreme heat and health

Copenhagen, Danemark

Oral Presentation

December, 2025

- Souilla L. (2025). In-situ responses of older adults living in social housing before, during and after a heatwave

Global School on Heat Adaptation

Montréal, Canada

Session animator

July 2025

- Souilla L. (2025). Demonstration and implementation of HeatSuite to monitor physiological and perceptual responses to heat

Seminars of the EPIC Center

Montréal, Canada

Oral Presentation

September 2025

- Souilla L. (2025). Deriving indoor heat-health relationships from remote physiological and behavioural monitoring of older adults living in social housing

Annual research days: School of Kinesiology and Physical Activity Sciences

Montréal, Canada

Poster Presentation

March 2025

- Souilla L. (2025). In situ responses of elderly people living in social housing before, during, and after a heatwave.

International Congress of Association for European Paediatric Congenital Cardiology

Porto, Portugal

Oral Presentation

May 2024

- Souilla L., Guillaumont S., ..., Amedro P. (2024). Cardiac rehabilitation in children and adolescents with long QT syndrome: the RYTHMO'FIT pilot study. International Congress of Association for European Paediatric Congenital Cardiology

Heart and sport days of Cardiogen

Montpellier, France

Oral Presentation

Jan. 2024

- Souilla L.(2024). Role of adapted physical activity in pediatric cardiac rehabilitation. Heart and sport days of Cardiogen

French federatation of Cardiology

Paris, France

Oral Presentation

Sep. 2023

- Souilla L., Werner O., ..., Amedro P. (2023). Health-related quality of life and physical activity in children with inherited cardiac arrhythmia or inherited cardiomyopathy: the prospective multicentre controlled QUALIMYORYTHM. Research days of French federation of cardiology

International Congress of Francophone Mother-Children network

Bordeaux, France

Oral presentation

June 2022

- Souilla L. (2022). Adapted physical activity teacher: on the right track. International Congress of Francophone Mother-Children network

French society of Paediatric

Lille, France

Oral presentation

June 2022

- Souilla L., Avesani M., Boisson A., Pasquie JL., Amedro P. (2022). Cardiorespiratory fitness and muscular deconditioning children with Long QT syndrome. French society of Paediatric

International Congress of Association for European Paediatric Congenital Cardiology

Geneve, Switzerland

Poster Presentation

May 2022

- Souilla L., Avesani M., Boisson A., Pasquie JL., Amedro P. (2022). Cardiorespiratory fitness and muscular deconditioning children with Long QT syndrome. International Congress of Association for European Paediatric Congenital Cardiology

Congress of French society of Paediatrics and congenital cardiology

Nice, France

Oral presentation

Sept. 2021

- Souilla L.(2021). Exercise rehabilitation in heart failure. Congress of French society of Paediatrics and congenital cardiology

Training & Schools

Summer Program in Aging School

Summer 2026

Canadian Institutes of Health Research (CIHR)

- Competitive national summer school on Impact of Climate Change on the Health of Older Persons; accepted as a local participant

Global School on Heat Adaptation *Summer 2025*

Global center on Heat Adaptation

- Engage students, stakeholders, and researchers from various disciplines globally involved in understanding and reducing the impacts of extreme heat (member of the team that organized the event).

Projects

Programming projects

Core Temp App	Analyze, plot and extract data from Body Cap pills used to measure body core temperature	*R*
CPET COSMED	Upload, analyze, plot result from COSMED cardiopulmonary exercise test and generate an individualized report	*Scilab*

Skills

Soft skills

Teamwork

Interdisciplinary Working – Effective communication – Autonomous writing – Management of national clinicals trials – Pedagogy with youth

Attributs

Creativity – Adaptability – Time management – Motivation – Relationship skills

Technical skills

Tools

CP Exercise testing – Climatic chamber – In-situ heat monitoring – Muscular ultrasound – Accelerometer – Force Sensor

Languages

French (native) – English (C1) – Spanish (B1)

Other

Github – Markdown – RStudio – Quarto – Shiny – Obsidian – Canva – BioRender

Memberships & Leadership

Oct. 2025-Nov. 2026	Creation and coordination of a student scientific journal club (Crush Your Brain)	*Daniel Gagnon Lab*
Oct. 2024-Oct. 2025	Scientific member of the international Global Center for Heat Adaptation	*Montreal Heart Institute*
Aug. 2023-Jan. 2025	Head of Green Researchers team working on evaluation and mitigation of the laboratory carbon footprint	*Lab Euromov DHM*
Jan. 2023-Dec. 2023	Head of Pediatric committee of the French Society of Exercise Physiologists	*French Society of Exercise Physiologists*
Since 2022	Scientific member of QUALINEURO-REHAB European project	*Bordeaux University Hospital - INSERM*